World Development Indicators Analysis (2022)

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1 Introduction

According to the World Bank [], the WDI dataset provides internationally comparable development indicators.

This report analyses a sample of the World Development Indicators (WDI) dataset for the year 2022. The dataset includes 14 variables across 217 countries. The goal is to conduct exploratory data analysis (EDA), create visualisations, and produce a reproducible report in both HTML and PDF formats.

2 Load Data

```
import pandas as pd

df = pd.read_csv("wdi.csv")
df.shape

df.head(5)
```

	country	inflation_rate	exports_gdp_share	gdp_growth_rate	gdp_per_capita	adult_lite
0	Afghanistan	NaN	18.380042	-6.240172	352.603733	NaN
1	Albania	6.725203	37.395422	4.856402	6810.114041	98.5
2	Algeria	9.265516	31.446856	3.600000	5023.252932	NaN
3	American Samoa	NaN	46.957520	1.735016	19673.390102	NaN
4	Andorra	NaN	NaN	9.563798	42350.697069	NaN

3 Exploratory Data Analysis

This section explores five key indicators:

- GDP per capita
- Life expectancy
- Inflation rate
- Unemployment rate
- Exports as a share of GDP

These indicators provide insight into economic performance, living standards, and macroeconomic conditions.

3.1 Select Indicators

```
selected = df[[
    "country",
    "gdp_per_capita",
    "life_expectancy",
    "inflation_rate",
```

```
    "unemployment_rate",
    "exports_gdp_share"
]]

selected.head()
```

	country	gdp_per_capita	life_expectancy	inflation_rate	unemployment_rate	exports_gd
0	Afghanistan	352.603733	62.879	NaN	14.100	18.380042
1	Albania	6810.114041	76.833	6.725203	11.588	37.395422
2	Algeria	5023.252932	77.129	9.265516	12.437	31.446856
3	American Samoa	19673.390102	NaN	NaN	NaN	46.957520
4	Andorra	42350.697069	NaN	NaN	NaN	NaN

3.2 Summary Statistics

```
selected.describe()
```

	gdp_per_capita	life_expectancy	inflation_rate	unemployment_rate	exports_gdp_share
count	203.000000	209.000000	169.000000	186.000000	169.000000
mean	20345.707649	72.416519	12.493936	7.268661	46.170395
std	31308.942225	7.713322	19.682433	5.827726	34.001404
min	259.025031	52.997000	-6.687321	0.130000	1.571162
25%	2570.563284	66.782000	5.518129	3.500750	24.526642
50%	7587.588173	73.514634	7.967574	5.537500	40.221277
75%	25982.630050	78.475000	11.665567	9.455250	55.460067
max	240862.182448	85.377000	171.205491	37.852000	211.278206

3.3 Missing Values

```
selected.isna().sum()
```

```
country               0
gdp_per_capita       14
life_expectancy       8
inflation_rate       48
```

```
unemployment_rate    31
exports_gdp_share    48
dtype: int64
```

The summary statistics show variation across countries in GDP per capita, inflation, and unemployment. Some indicators contain missing values, which is expected in international datasets due to differences in data availability.

4 Visualisations

4.1 Relationship between GDP per capita and Life Expectancy

```python
import matplotlib.pyplot as plt

plt.figure(figsize=(8,6))
plt.scatter(selected["gdp_per_capita"], selected["life_expectancy"])

plt.xlabel("GDP per capita (constant US$)")
plt.ylabel("Life expectancy (years)")
plt.title("GDP per capita vs Life Expectancy")

plt.show()
```

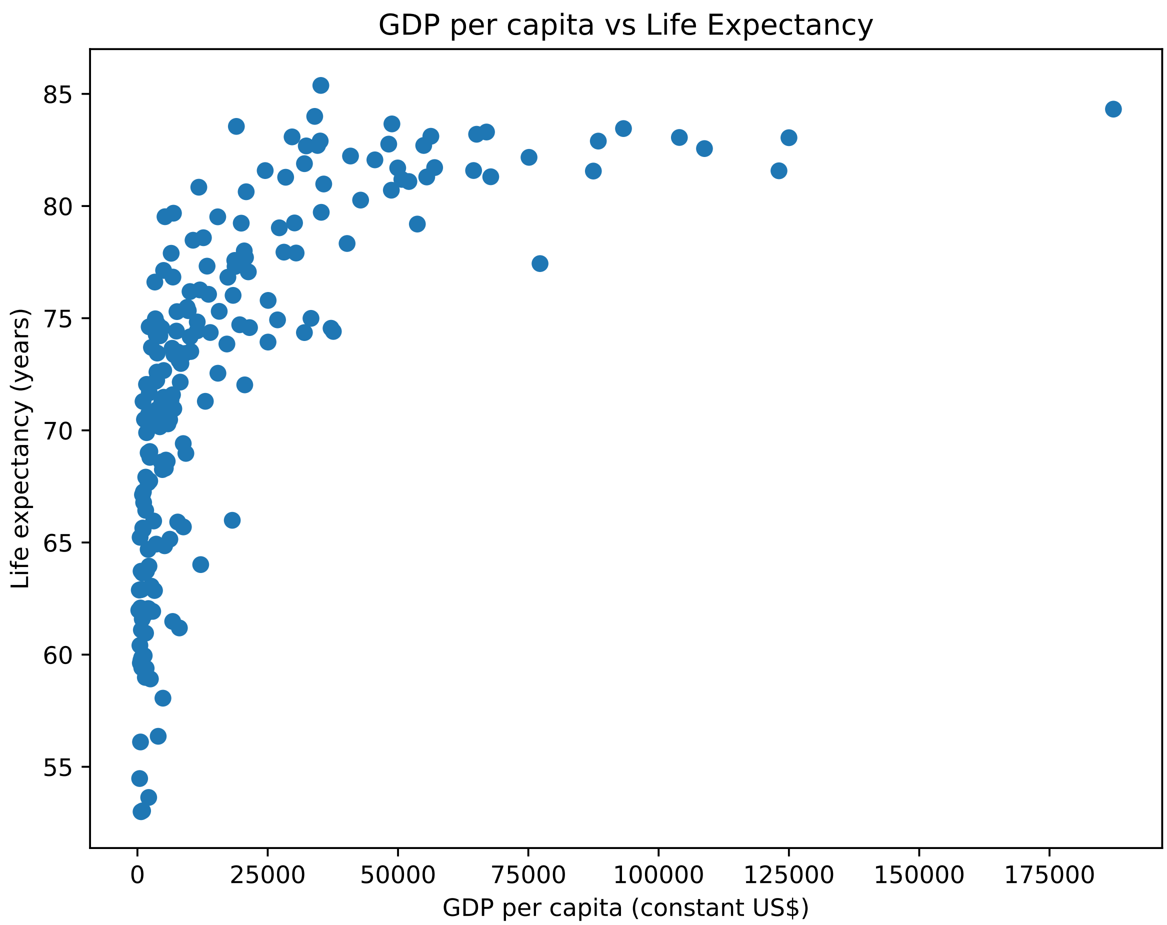

Figure 1: Relationship between GDP per capita and life expectancy (World Bank, 2022)

Figure Figure 1 shows a positive relationship between GDP per capita and life expectancy.

4.2 Top 10 Countries by GDP per Capita

```
top10 = selected.sort_values("gdp_per_capita", ascending=False).head(10)

plt.figure(figsize=(10,6))
plt.bar(top10["country"], top10["gdp_per_capita"])

plt.xlabel("Country")
plt.ylabel("GDP per capita (constant US$)")
```

```
plt.title("Top 10 Countries by GDP per Capita")

plt.xticks(rotation=45)

plt.show()
```

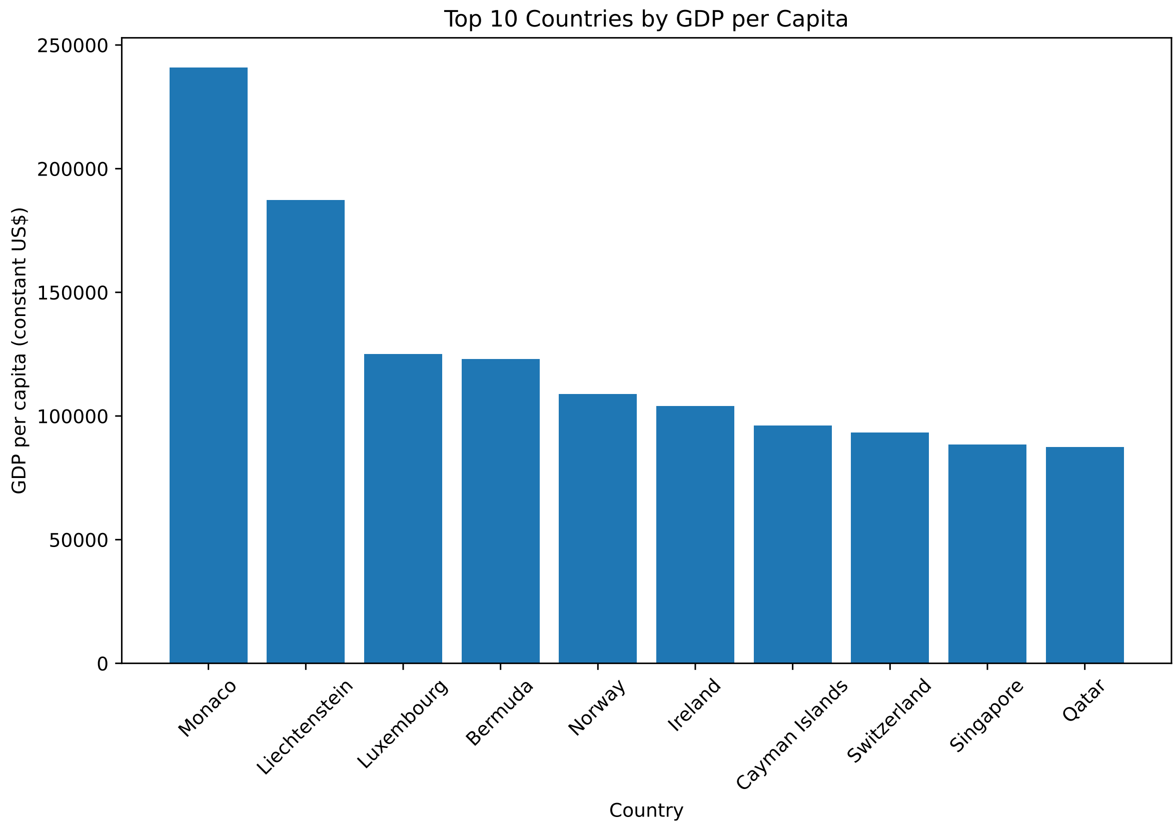

Figure 2: Top 10 countries by GDP per capita (World Bank, 2022)

Figure Figure 2 displays the countries with the highest GDP per capita. # Summary Statistics Table

The key descriptive statistics for the selected indicators are shown in Table Table 4.

```
summary_table = selected[[
    "gdp_per_capita",
    "life_expectancy",
    "inflation_rate",
```

```
    "unemployment_rate",
    "exports_gdp_share"
]].describe()

summary_table
```

Table 4: Summary statistics of selected development indicators (World Bank, 2022)

	gdp__per__capita	life__expectancy	inflation__rate	unemployment__rate	exports__gdp__share
count	203.000000	209.000000	169.000000	186.000000	169.000000
mean	20345.707649	72.416519	12.493936	7.268661	46.170395
std	31308.942225	7.713322	19.682433	5.827726	34.001404
min	259.025031	52.997000	-6.687321	0.130000	1.571162
25%	2570.563284	66.782000	5.518129	3.500750	24.526642
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max	240862.182448	85.377000	171.205491	37.852000	211.278206

Table Table 4 shows variation across countries in economic and social indicators. GDP per capita and inflation show substantial variability, reflecting differences in economic development and macroeconomic stability across countries.

Bank, World. 2022a. "World Bank Data Methodology." https://data.worldbank.org.

———. 2022b. "World Development Indicators." https://databank.worldbank.org/source/world-development-indicators.